Yakeen Plus (2025)

SHORT PRACTICE TEST - 04

DURATION : 60 Minutes | DATE : 06/10/2024 | M. MARKS : 192

Answer Key

PHYSICS		CHEMISTRY		BOTANY		ZOOLOGY	
1.	(4)	13.	(3)	25.	(3)	37.	(1)
2.	(2)	14.	(1)	26.	(3)	38.	(3)
3.	(1)	15.	(1)	27.	(2)	39.	(4)
4.	(3)	16.	(2)	28.	(3)	40.	(4)
5.	(4)	17.	(1)	29.	(3)	41.	(1)
6.	(1)	18.	(4)	30.	(4)	42.	(4)
7.	(2)	19.	(1)	31.	(4)	43.	(4)
8.	(1)	20.	(3)	32.	(2)	44.	(4)
9.	(2)	21.	(1)	33.	(4)	45.	(3)
10.	(3)	22.	(1)	34.	(3)	46.	(4)
11.	(2)	23.	(4)	35.	(4)	47.	(1)
12.	(2)	24.	(1)	36.	(4)	48.	(3)

SECTION-(I) PHYSICS

1. **(4)**

$F \propto \frac{1}{r^2}$. If r becomes double then F reduces to $\frac{F}{4}$

[New NCERT Class 11th Page No. 130]

2. **(2)**

The value of g at the height h from the surface of earth

$$g' = g\left(1 - \frac{2h}{R}\right)$$

The value of g at depth x below the surface of earth

$$g' = g\left(1 - \frac{x}{R}\right)$$

These two are given equal,

$$\therefore \left(1 - \frac{2h}{R}\right) = \left(1 - \frac{x}{R}\right)$$

On solving, we get

$x = 2h$

[New NCERT Class 11th Page No. 135]

3. **(1)**

Escape velocity is same for all angles of projection.

[New NCERT Class 11th Page No. 136]

4. **(3)**

$$l = \frac{FL}{AY}$$

$$\Rightarrow l \propto \frac{L}{d^2}$$

$$\Rightarrow \frac{l_1}{l_2} = \frac{L_1}{L_2} \times \left(\frac{d_2}{d_1}\right)^2$$

$$= \frac{1}{2} \times \left(\frac{1}{2}\right)^2 = \frac{1}{8}$$

[New NCERT Class 11th Page No. 171]

5. **(4)**

$$\frac{Y_A}{Y_B} = \frac{\tan\theta_A}{\tan\theta_B} = \frac{\tan 60}{\tan 30} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

$$\Rightarrow Y_A = 3Y_B$$

[New NCERT Class 11th Page No. 170]

6. **(1)**

$$l = \frac{FL}{AY}$$

$\therefore l \propto \frac{1}{r^2}$ (Y, L and F are constant)

i.e. for the same load, thickest wire will show minimum elongation. So graph D represent the thickest wire.

[New NCERT Class 11th Page No. 169]

7. **(2)**

Speed of the earth will be maximum when its distance from the sun is minimum because mvr = constant

[New NCERT Class 11th Page No. 129]

8. **(1)**

From the graph $l = 10^{-4}$m, $F = 20$ N

$A = 10^{-6}$ m^2, $L = 1$ m

$$\therefore Y = \frac{FL}{Al}$$

$$= \frac{20 \times 1}{10^{-6} \times 10^{-4}} = 20 \times 10^{10} = 2 \times 10^{11}\ \text{N/m}^2$$

[New NCERT Class 11th Page No. 185]

9. **(2)**

Let h be the depth of lake and P_0 be the atmospheric pressure.

According to the question,

Pressure at half depth of lake

$= \frac{2}{3} \times$ (Pressure at bottom of lake)

$$P_0 + \rho g\frac{h}{2} = \frac{2}{3}(\rho gh + P_0)$$

$$\Rightarrow 3\left(P_0 + \rho g\frac{h}{2}\right) = 2(\rho gh + P_0)$$

$$\left[\because P_0 = 10^5\,\text{Pa},\ \rho = 10^3\,\text{kg/m}^3\right]$$

$$\Rightarrow P_0 + \frac{3}{2}\rho gh = 2\rho gh + 2P_0$$

$$\Rightarrow P_0 = \left(2 - \frac{3}{2}\right)\rho gh$$

$$\Rightarrow P_0 = \frac{\rho gh}{2}$$

$$\Rightarrow h = \frac{2P_0}{\rho g} = \frac{2 \times 10^5}{10^3 \times 10} = 20\,\text{m}$$

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10. **(3)**

Applying equation of continuity at the junction of three tubes,

$A_1v_1 = A_2v_2 + A_3v_3$ ....(i)

where, $A_1 = A, A_2 = A, A_3 = 1.5\ A$

and $v_1 = 3$ m/s, $v_2 = 1.5$ m/s, $v_3 = v$

Substituting in Eq.(i),

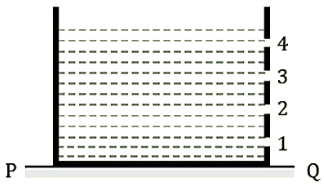
$Av_1 = Av_2 + 1.5\ Av$

Or, $A \times 3 = (A \times 1.5) + (1.5\ Av)$

$\therefore\ v = 1$ m/s

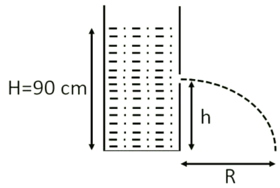
[New NCERT Class 11th Page No. 188]

11. **(2)**



We know that for maximum horizontal range, height of the hole from the floor is given by

$$h = \frac{H}{2}$$



$$\Rightarrow h = \frac{90}{2} = 45\,\text{cm} = h_3$$

Therefore, water coming out from hole 3 will fall on the ground at maximum horizontal range.

[New NCERT Class 11th Page No. 187]

12. **(2)**

$$\frac{1}{R} = \frac{1}{r_1} - \frac{1}{r_2}$$

$$\Rightarrow R = \frac{r_1 r_2}{r_2 - r_1}$$

$$R = \frac{5 \times 4}{5-4} = 20\ \text{cm.}$$

[New NCERT Class 11th Page No. 194]

SECTION-(II) CHEMISTRY

13. **(3)**

$X\,|\,X^{2+}(0.1M)\,||\,Y^{2+}(1M)\,|\,Y$

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

$$= 0.34 - (-0.25)$$

$$= 0.59V$$

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n}\log\frac{[X^{2+}]}{[Y^{2+}]}$$

$$E_{cell} = 0.59 - \frac{0.059}{2}\log\frac{0.1}{1}$$

$$= 0.59 + \frac{0.059}{2}$$

$$= 0.62\,V$$

[New NCERT Class 12thPage No. 35]

14. **(1)**

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF}\ln\frac{[Ni^{2+}]}{[Cu^{2+}]}$$

[New NCERT Class 12th Page No. 38]

15. **(1)**

$SrCl_2$ will exhibit the maximum value of equivalent conductance in the fused state. On moving down the group, the size of an element increases, the covalent character will eventually decrease and hence the ionic character increases. Thus, $SrCl_2$ will be most ionic and thus will exhibit maximum equivalent conductance in the fused state as it will readily gives ions.

[New NCERT Class 12th Page No. 43]

16. **(2)**

Amount of Ag in solution = M × V (mL)

= 1 × 250 = 250 m mol

= 250×10^{-3} mol

= 0.25 mol

$Ag^+ + e^- \rightarrow Ag$

Reduction of 1 mol of Ag requires 96500 C

Reduction of 0.25 mol of Ag requires

= 96500 × 0.25

= 24125 C

[New NCERT Class 12th Page No. 52]

17. **(1)**

To obtain a straight-line graph, we need to manipulate the equation into a form representing a linear relationship (y = mx + c).

The given equation is:

$$Rt = \log C_0 - \log C_t$$

$$t = \frac{1}{R}\log C_0 - \frac{1}{R}\log C_t$$

Hence, the graph between time (t) and log C_t will given a straight line with negative slop of (–1/R) and the intercept will be log C_0/R.

[New NCERT Class 12th Page No. 74]

18. (4)

$$10^{15}e^{-1000/T} = 10^{16}e^{-2000/T}$$

$$\frac{e^{-2000/T}}{e^{-1000/T}} = 10^{-1}$$

$$e^{-1000/T} = 10^{-1}$$

$$\log_e e^{-1000/T} = \log_e 10^{-1}$$

$$\frac{-1000}{T} = 2.303\ \log_{10}10^{-1} = 2.303$$

$$T = \frac{1000}{2.303}K$$

[New NCERT Class 12th Page No. 79]

19. (1)

From the observation, we can see that changing B has no effect on the rate, and doubling the amount of the A doubles the rate of the reaction.

Thus, Rate = $k[A]^1$ satisfies the given data.

Thus order = 1

[New NCERT Class 12th Page No. 72]

20. (3)

For endothermic reaction, activation energy is greater than enthalpy change in the reaction.

[New NCERT Class 12th Page No. 79]

21. (1)

Bohr's atomic theory is applicable to hydrogen and monoatomic lons like

He^{+}, Li^{2+}, Be^{3+}, etc.

[New NCERT Class 11th Page No. 49]

22. (1)

We know that

$$\frac{1}{\lambda_{min}} = R\left(\frac{1}{4}-\frac{1}{\infty}\right) = \frac{R}{4};$$

$$\frac{1}{\lambda_{max}} = R\left(\frac{1}{4}-\frac{1}{9}\right)$$

$$= \frac{5R}{36} \quad \Rightarrow \quad \frac{\lambda_{min}}{\lambda_{max}} = \frac{\frac{4}{R}}{\frac{36}{5R}} = \frac{5}{9}$$

[New NCERT Class 11th Page No. 48]

23. (4)

$$E = \frac{hc}{\lambda}$$

$$E \propto \frac{1}{\lambda}$$

Inverse relation shows that there will be rectangular hyperbola, in the plot of E against λ.

[New NCERT Class 11th Page No. 41]

24. (1)

- Option (i) is correct
- Orbital is the space in which the probability of finding the electron is maximum $(\approx 95\%)$.
- The probability of finding an electron in an orbital can not be zero.

[New NCERT Class 11th Page No. 34]

SECTION-(III) BOTANY

25. (3)

The Mycoplasma are organisms that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen.

[New NCERT Class 11th Page No. 14]

26. (3)

Protozoans are included in kingdom Protista.

[New NCERT Class 11th Page No. 15, 16]

27. (2)

Diatoms are microscopic.

[New NCERT Class 11th Page No. 14]

28. (3)

- Organisms belongs to kingdom animalia follow a definite growth pattern.
- Higher forms of animalia kingdom shows elaborate sensory and neuromotor mechanism.

[New NCERT Class 11th Page No. 19]

29. (3)

In basidiomycetes, the mycelium is branched and septate.

[New NCERT Class 11th Page No. 17, 18]

30. (4)

Ustilago	Smut
Puccinia	Rust
Agaricus	Mushroom
Albugo	Parasitic fungi on mustard

[New NCERT Class 11th Page No. 17, 18]

31. (4)

- Slime moulds are saprophytic protists.
- Euglenoids-Majority of them are fresh water organisms found in stagnant water.
- Diatoms-Float passively in water currents (plankton). Flagella absent in diatoms.

[New NCERT Class 11th Page No. 15]

32. **(2)**
During unfavourable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips. The spores possess true walls.
[New NCERT Class 11th Page No. 15]

33. **(4)**
In basidiomycetes, karyogamy and meiosis take place in the basidium producing four basidiospores.
[New NCERT Class 11th Page No. 18]

34. **(3)**

Viroids	Potato spindle tuber disease
Bacteriophage	Double stranded DNA
Prions	Mad cow disease in cattle
Tobacco Mosaic Virus	Single stranded RNA

[New NCERT Class 11th Page No. 20, 21]

35. **(4)**
Bacteria adopt sexual reproduction by primitive type of DNA transfer. Mushrooms, bracket fungi and puffballs are commonly known forms of basidiomycetes.
[New NCERT Class 11th Page No. 14, 17, 18]

36. **(4)**
Sporozoans: This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is *Plasmodium* (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.
[New NCERT Class 11th Page No. 16]

SECTION-(IV) ZOOLOGY

37. **(1)**
Urine is stored in the urinary bladder until a voluntary signal comes from the central nervous system. The receptors present on the urinary bladder receive these signals.
[New NCERT Class 11th Page No. 240]

38. **(3)**
- Renin converts angiotensinogen into angiotensin I.
- An increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF).

[New NCERT Class 11th Page No. 214]

39. **(4)**
The process of release of urine is called micturition and the neural mechanisms causing it is called micturition reflex. An adult human excretes about 1 to 1.5 litres of urine per day.
[New NCERT Class 11th Page No. 241]

40. **(4)**
Vasa recta is well developed in juxta medullary nephrons.
[New NCERT Class 11th Page No. 214]

41. **(1)**
The fluid filtered from the glomerular capillaries into the Bowman's capsule is equivalent to plasma without proteins.
[New NCERT Class 11th Page No. 208]

42. **(4)**
Presence of glucose (Glycosuria) and ketone bodies (Ketonuria) in urine are indicative of diabetes mellitus.
[New NCERT Class 11th Page No. 213]

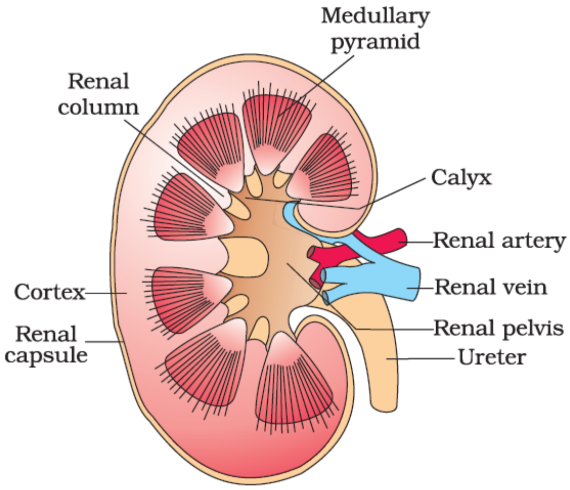
43. **(4)**

Ammonotelic	Bony fishes
Uricotelic	Reptiles
Ureotelic	Marine fishes
Bowman's capsule	Renal tubule

[New NCERT Class 11th Page No. 242]

44. **(4)**
Ascending limb of loop of Henle is impermeable to water and allow transport of electrolyte actively and passively.
[New NCERT Class 11th Page No. 244]

45. **(3)**



[New NCERT Class 11th Page No. 202]

46. **(4)**
Reabsoption of water from distal parts of tubule is facilitated by Vasopression or ADH hormone.
[New NCERT Class 11th Page No. 212]

47. (1)

- Excessive loss of body fluid from the body switch on the osmoreceptor.
- ADH is a vasoconstrictor.
- ADH is responsible for the increase in GFR.

[New NCERT Class 11^{th} Page No. 212]

48. (3)

Urethral sphincter causes the release of urine to the outside of the body. This process starts when the central nervous system passes on the motor messages to the smooth muscles of the bladder. This signal initiates their contraction. Simultaneously, relaxation of the urethral sphincter takes place which causes the release of the urine.

[New NCERT Class 11^{th} Page No. 247]

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